

HAT-P-36b

R-0721

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260315 · created 2026-07-27T23:58:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461114.8074 +/- 0.0055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.029
Transit depth (Rp/Rs)^2	1.03 +/- 0.59 [%]
Orbital Inclination (inc)	80.22 +/- 2.88
Airmass coefficient 1 (a1)	1.268 +/- 0.015
Airmass coefficient 2 (a2)	-0.22 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	3.83
Optimal Annulus	11.71
Transit Duration (day)	0.064 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.102 ± 0.029 vs 0.126 (deviation 0.75σ)
Duration fit vs published	0.064 d vs 0.0925 d (deviation 1.07σ)
Mid-time O–C	-16.7 min
Depth SNR	3.2
Residual scatter	1.22 %

Light curve

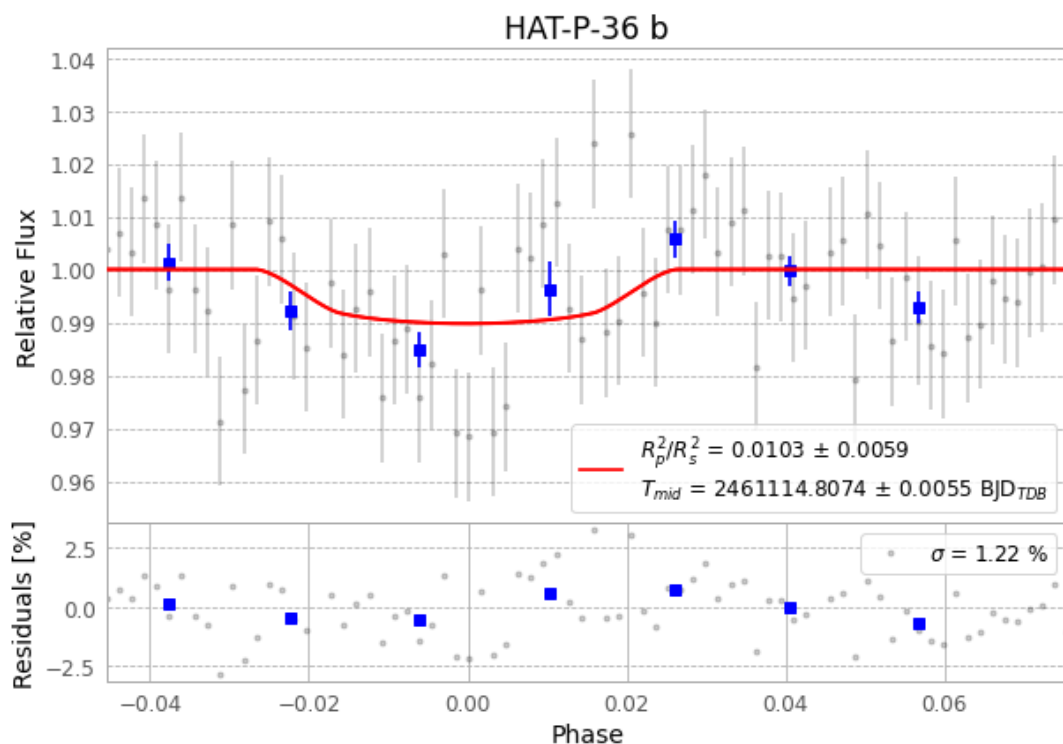


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-03-14.png (SHA-256 98737994ab969180...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 108], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a6309adebbaef256...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), HATP-36260315054815.FITS → HATP-36260315093915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260315.log (c592819d00ce...), FinalLightCurve_HAT-P-36 b_2026-03-14.png (98737994ab96...), FinalLightCurve_HAT-P-36 b_2026-03-14.csv (7340a26191c2...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:58Z.